WORKSHEET 5.1

The Mean Value Theorem

Unit 5: Analysing Functions

Key ideas

- **Mean Value Theorem.** If f is continuous on $[a, b]$ and differentiable on (a, b), there is a c in (a, b) with $f'(c) = \dfrac{f(b) - f(a)}{b - a}$.
- In words: at some point the instantaneous rate equals the average rate.
- If also $f(a) = f(b)$, then $f'(c) = 0$ for some c (Rolle's Theorem).
- To *use* the theorem in a justification, state both conditions. "Differentiable, and therefore continuous" covers both.

Worked examples

Example 1 · Finding c

Find the value of c guaranteed by the Mean Value Theorem for $f(x) = x^2 - 2x$ on $[1, 4]$.

Solution. $\dfrac{f(4) - f(1)}{3} = \dfrac{8 - (-1)}{3} = 3$. Solve $2c - 2 = 3$.

$c = \dfrac{5}{2}$.

Example 2 · A reciprocal

Find the value of c guaranteed by the theorem for $f(x) = \dfrac{1}{x}$ on $[1, 9]$.

Solution. $\dfrac{\frac{1}{9} - 1}{8} = -\dfrac{1}{9}$. Solve $-\dfrac{1}{c^2} = -\dfrac{1}{9}$ with $1 < c < 9$.

$c = 3$.

Example 3 · From values

A differentiable function g has $g(0) = 5$ and $g(4) = 13$. What does the theorem guarantee?

Solution. $\dfrac{13 - 5}{4} = 2$, and g is differentiable, hence continuous, on $[0, 4]$.

There is a c in $(0, 4)$ with $g'(c) = 2$.

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Practice

CORE

1. Find the value of c guaranteed by the Mean Value Theorem for $f(x) = x^2 + 1$ on $[0, 2]$.

2. Find the value of c guaranteed by the theorem for $f(x) = x^3$ on $[-1, 2]$.

3. State the two conditions of the Mean Value Theorem.

4. Explain why the theorem does not apply to $f(x) = |x|$ on $[-1, 1]$.

EXAM LEVEL

5. Find the value of c guaranteed by the theorem for $f(x) = \sqrt{x}$ on $[1, 9]$.

6. Find the value of c guaranteed by the theorem for $f(x) = \ln x$ on $[1, e]$.

7. A car travels 150 km along a road in exactly 2 hours. Explain why its speedometer must have read 75 km/h at some instant. State any assumption you make.

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8. A differentiable function h has $h(2) = 3$ and $h(7) = 3$. Must there be a c in $(2, 7)$ with $h'(c) = 0$? Must there be a c with $h(c) = 0$?

Question 9 · Free-response practice

Let $f(x) = x^3 - x^2$.

(a) Explain why the Mean Value Theorem applies to f on $[0, 2]$.

(b) Find the value of c in $(0, 2)$ guaranteed by the theorem.

(c) Show that $f(0) = f(1)$, and find the value of c in $(0, 1)$ at which $f'(c) = 0$.

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Worked solutions

1. Average rate $\dfrac{5-1}{2} = 2$; $2c = 2$ gives $c = 1$.

2. Average rate $\dfrac{8-(-1)}{3} = 3$; $3c^2 = 3$ gives $c = 1$ (the value $c = -1$ is not inside the interval).

3. f is continuous on $[a, b]$ and differentiable on (a, b).

4. f is not differentiable at $x = 0$, which lies in $(-1, 1)$.

5. Average rate $\dfrac{3-1}{8} = \dfrac{1}{4}$; $\dfrac{1}{2\sqrt{c}} = \dfrac{1}{4}$ gives $c = 4$.

6. Average rate $\dfrac{1-0}{e-1}$; $\dfrac{1}{c} = \dfrac{1}{e-1}$ gives $c = e - 1$.

7. Assuming the position is a differentiable function of time, its average rate of change is $\dfrac{150}{2} = 75$ km/h, so by the Mean Value Theorem the velocity equals 75 km/h at some instant.

8. Yes to the first: h is differentiable and $h(2) = h(7)$, so the theorem gives $h'(c) = \dfrac{3-3}{5} = 0$. No to the second: nothing forces h to take the value 0 (for example h could be constant at 3).

9. (a) f is a polynomial, so it is continuous on $[0, 2]$ and differentiable on $(0, 2)$.

   (b) $\dfrac{f(2) - f(0)}{2} = \dfrac{4}{2} = 2$. $3c^2 - 2c = 2$ gives $c = \dfrac{1+\sqrt{7}}{3} \approx 1.215$ (the other root is negative).

   (c) $f(0) = 0$ and $f(1) = 1 - 1 = 0$. $f'(c) = 3c^2 - 2c = c(3c - 2) = 0$ in $(0, 1)$ gives $c = \dfrac{2}{3}$.

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